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Effects of Baduanjin exercise on cancer-related fatigue in patients with prostate cancer treated with androgen deprivation therapy in Shanghai, China: a study protocol for a randomised controlled trial

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ABSTRACT

Introduction Cancer-related fatigue (CRF) is one of the most common and painful symptoms in patients with prostate cancer (PCa). Moreover, PCa patients who receive the androgen deprivation therapy (ADT) are more likely to develop CRF. Baduanjin exercise has been shown to improve CRF in some cancers. However, such effects have not been verified in patients with PCa treated with the ADT. So, this study was designed as a randomised controlled trial (RCT) to explore the effects of Baduanjin exercise on CRF in PCa patients treated with the ADT.

Methods and analysis This study will be a single-centre, assessor and statistician blinded, RCT consisting of a 12 week intervention and 12 week follow-up. Patients with PCa who meet the inclusion criteria will be recruited from Shanghai Hudong Hospital. Participants will be randomly assigned to Baduanjin exercise group (n=42) and the control group (n=42), performing 12 weeks of Baduanjin exercise or reeiving the standard care. The primary outcome will be the clinical effect of Baduanjin exercise on CRF in PCa patients, which will be measured using the Piper Fatigue Self-Assessment (PFS) scale, a multidimensional measure of CRF using three different dimensions: somatic, emotional and cognitive. The secondary outcome will be the clinical effect of the Baduanjin exercise on the patient's level of fatigue, sleep, depression and life quality at the time, which will be assessed by the Brief Fatigue Inventory (BFI), the Pittsburgh Sleep Quality Index (PSQI), the Beck Depression Inventory (BDI) and the Functional Assessment of Cancer Therapy-Prostate (FACT-P).

Ethics and dissemination This study has been approved by Shanghai Hudong Hospital Ethics Committee, Shanghai province (2022 SHHDKY08). The trial results will be submitted to conferences and peer-reviewed journals.

Trail registration number ChiCTR2300074293

STRENGTHS AND LIMITATIONS OF THIS STUDY

- ⇒ This study will be a single-centre, rigorously randomised, assessor and statistician blinded, randomised clinical controlled trial consisting of a 12 week intervention and 12 week follow-up.
- ⇒ This trial will use comprehensive outcome measures, both multidimensional and uni-dimensional, to assess the extent of cancer-related fatigue (CRF), as well as patients' sleep, depression and quality of life, which will support the clinical efficacy of Baduanjin exercise for CRF in prostate cancer patients treated with androgen deprivation therapy.
- ⇒ All participants will agree to register with a unique identification number and provide a mobile phone number to facilitate follow-up.
- ⇒ Due to the visible nature of the exercise intervention, it is difficult to ensure that the researchers and participants are blinded.

INTRODUCTION

Prostate cancer (PCa) is the second most prevalent cancer among men globally, posing a significant threat to their overall health.[1–3] In the current treatment programme, androgen deprivation therapy (ADT) is a primary treatment option for PCa,[4] as it suppresses the production of testosterone/androgens. However, ADT treatment often leads to significant side effects,[5] including cancer-related fatigue (CRF), altered body composition (decreased lean mass and increased fat mass) and increased risk of cardiovascular disease, insulin resistance and metabolic syndrome.[6] Therefore, recent clinical focus has shifted towards managing or reducing ADT-related side effects.[4 7–9] CRF is the most commonly encountered side effect in PCa patients treated with ADT, with a prevalence of approximately 74%.[10] Through in-depth qualitative research, it was found that CRF interferes with patients' basic

activities of daily living (ie, preparing food, driving and cleaning) and socialising and reduces treatment adherence.[11] It has even been reported that within 6 years post-treatment with ADT, one-third of PCa patients develop clinical symptoms of CRF.[12] Therefore, the reduction of CRF symptoms in PCa patients during ADT treatment has become an important scientific issue being addressed by researchers worldwide.

Exercise during ADT treatment is effective in counteracting CRF,[13 14] possibly by lowering the inflammatory activity and/or by increasing physical fitness.[15] However, at present, there is still a lack of evidence that PCa patients undergoing ADT can use moderate-intensity exercise prescriptions. International guidelines recommend three sessions of at least moderate-intensity endurance training and/or two sessions of at least moderate-intensity resistance training each week to counteract CRF.[16] The guideline suggests that physical exercise and psychosocial interventions can be used to treat CRF.[17] And there are meta-analyses suggesting that exercise therapy can improve CRF in patients,[14 18] especially the exercises that focus on both mind and body therapy.[14 17] However, the evidence base regarding the ideal exercise prescription in terms of exercise type, frequency, intensity and duration for cancer survivors is insufficient.[16] While low-to-moderate intensity may be preferred by most patients, one study found that moderate-to-high-intensity exercise during cancer treatment was beneficial for physical fatigue compared with low-intensity exercise.[19]

The Baduanjin exercise is a moderate-intensity aerobic exercise that originated in China and is widely used in Asian countries for daily exercise.[20] In recent years, the benefits of Baduanjin have been demonstrated in the improvement of CRF in a variety of cancer patients.[21] As an exercise therapy in Traditional Chinese Medicine, not only does the Baduanjin exercise help improve the body's digestive, respiratory, circulatory and locomotor functions, it also emphasises the importance of physical and mental balance.[22 23] A study strongly suggests that the Baduanjin relieves CRF in patients with colorectal cancer undergoing chemotherapy and also improves the level of physical activity and sleep quality.[24] A systematic review and meta-analysis showed that Baduanjin exercise can improve the quality of life and alleviate anxiety and depressive state among postoperative patients with breast cancer.[25] A recent study suggested that cognitive–behavioral therapy combined with Baduanjin exercise can improve fatigue and cognitive impairment of colorectal cancer patients undergoing chemotherapy and improve their quality of life.[21] Consisting of eight routinely conducted sections of movements,[26] Baduanjin exercise is a combination of breathing, mind and body movement fitness methods. Compared with modern exercises, Baduanjin exercise focuses more on improving body strength and psychological construction by developing balanced maintenance of the body and mind,[22 23] few other types of exercise involve breathing and meditation in the process.[26] However, it is still unknown whether Baduanjin exercise can be a potential exercise prescription for patients receiving ADT.

In light of this, this randomised controlled trial (RCT) aimed to investigate the efficacy and safety of the Baduanjin exercise in improving CRF in PCa patients treated with ADT. We expect this study to provide high-quality clinical evidence for using the Baduanjin exercise in treating CRF in PCa patients undergoing ADT.

METHODS AND MATERIALS

Study design

The study will be conducted as a single-centre, assessor and statistician blinded, RCT based on Consolidated Standards of Reporting Trials. The subjects will be patients with PCa treated with ADT in Shanghai Hudong Hospital. There will be two parallel control groups in this study, and participants will be randomly assigned to Baduanjin exercise group (n=42) and the control group (n=42) in a 1:1 ratio. Participants in the Baduanjin exercise group will receive the Baduanjin exercise for 12 weeks,[18] whereas participants in the control group will receive the standard care.[7 27] The flow chart of the study is shown in figure 1. The schedules of trial enrolment, interventions and assessments are shown in table 1.

Study setting

The study will be conducted at Shanghai Hudong Hospital.

Safety measurements

During the intervention, the research assistants will monitor the adverse events (AEs) that occurred in the participants. The research assistants will report any AEs, such as exercise injuries and falls, to the researchers in detail using the AE case report form. The researchers will assess the causality of AEs associated with Baduanjin intervention and the severity of AEs. AEs will be reported to the ethics committee.

Sample size calculation

This study will be an RCT with a parallel design. The intervention group is the Baduanjin exercise group, and the control group is the standard care group, with the total score of PFS as the primary outcome indicator observed. The expected mean difference in the PFS score was approximately 1.4 between the intervention group and control group, based on the current related study.[28] Hence, with a power of 90% and type I error of 5% (two-sided), a sample size of 68 (34 per group) was calculated by the PASS 15 software. The final sample size will be 84 (42 per group), considering a 20% dropout rate.[29]

Inclusion and exclusion criteria

Inclusion criteria

1. Clinical diagnosis of PCa and treated with ADT for ≥3 months.[7]
2. PFS score ≥4 (moderate to severe fatigue).[17]
3. Can communicate and write normally.

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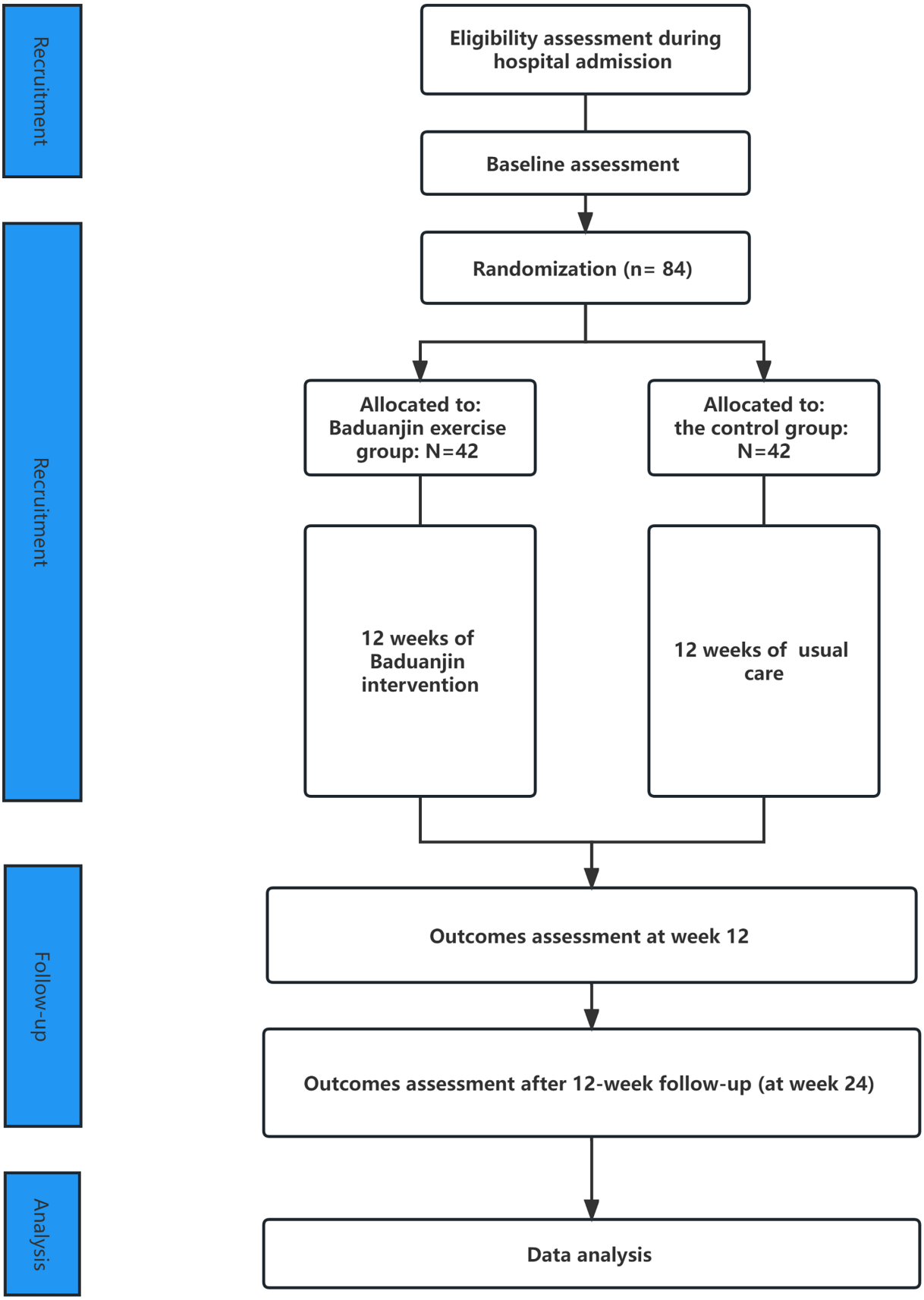

Figure 1 Consolidated Standards of Reporting Trial flow chart of the study.

Table 1 The schedule of trial enrolment, interventions and assessment

Timepoint	On enrolment (week 0)	Intervention period (week 1–12)	End of intervention (week 12)	Follow-up period (week 13–24)	End of follow-up (week 24)
Inclusion/exclusion criteria	×				
Informed consent	×				
Demographic characteristics	×				
Randomisation and allocation	×				
PFS	×		×		×
BFI	×		×		×
PSQI	×		×		×
BDI	×		×		×
FACT-P	×		×		×
Safety measurement		×	×	×	×

BDI, Beck Depression Inventory; BFI, Brief Fatigue Inventory; FACT-P, Functional Assessment of Cancer Therapy-Prostate; PFS, Piper Fatigue Self-Assessment Scale; PSQI, Pittsburgh Sleep Quality Index.

4. Aged between 18 and 70 years.
5. Voluntary participation, have understood and signed the written informed consent form.

Exclusion criteria

1. Have serious damage to the heart, liver, kidney and other important organs that may contraindicate Baduanjin exercise (eg, severe heart failure, severe chronic obstructive pulmonary disease or orthopaedic conditions).
2. Have any diseases affecting motor nerves, muscles and joints.
3. Unable to perform Baduanjin exercise due to physical conditions or other reasons.
4. Have any neoplastic cancer that may cause the CRF will be excluded (excluding basal cell skin carcinoma).
5. Have a mental illness or severe cognitive impairment or language barrier.
6. Participating in other clinical exercises or regularly exercising more than three times a week.
7. Expectation of life less than 1 year.
8. For other reasons, the researcher does not consider them suitable for inclusion.

Drop-out criteria

1. Serious adverse reactions in the process of the study.
2. Poor compliance or the amount of exercise is not up to standard (<80% or >120%).[30]
3. Conducting other clinical exercises without permission, affecting the efficacy and safety evaluation.
4. Quit, lost or died due to various reasons.

Recruitment

Participants will be recruited from Shanghai Hudong Hospital. Specifically, patients who meet the requirements will be recruited in the inpatient department and outpatient department of the hospital. In addition, the recruitment will be conducted through online advertisements, such as WeChat, etc. Inform potential participants of the purpose, procedure, intervention, potential adverse effects and advantages of the trial. During the recruitment process, two researchers will conduct data collection and telephone follow-up. To ensure the quality of data collection, two researchers will be trained in questionnaire data collection skills, including understanding questionnaire items and standardising conversations with participants. Each qualified participant is required to sign an informed consent form, provided that the potential participant met the inclusion criteria and volunteered to participate in the trial. The researchers of this study will not exclude patients because of their economic or social status. And each potential participant will be informed that they can withdraw from the trial at any time without consequence. Research assistants will be responsible for obtaining informed consent.

Randomisation and allocation concealment

84 participants will be randomly assigned to the Baduanjin exercise group or the control group in a 1:1 ratio. The randomly assigned sequence will be generated by an independent statistician through the random number generator (Statistics version 21.0 software; IBM Corp., Armonk, NY, USA) and using sequentially numbered opaque and sealed envelopes to achieve distribution hiding.

Blinding

Due to the visible nature of the exercise intervention, it is hard to ensure that the researchers and participants are blinded. Therefore, the blind method will be only guaranteed for the assessors and statisticians in this RCT. They will be responsible for data collection and analysis and

Table 2 Details of the operation of theaside to make Baduanjin exercise

Sequence of actions	Action details
Preparatory posture	Step your left foot aside to make the feet shoulder-width apart. Bend the knees and squat, put your palms against the anterior abdomen and calm down.
Action 1	*Hold up the hands* Separate the feet to the shoulder width, hold the chest and abdomen and look at the front horizontally. Calm down and regulate the breath, sink your breath to the lower abdomen, raise your hands overhead, turn the palms upward and lift hard and then raise the heel. Lift when raising the hands, hold when laying down the hands six times, turn the palms downward and press the lower abdomen.
Action 2	*Draw a bow like shooting a hawk* Separate the legs and bend the knees, draw the hands up and pull as if drawing a bow to the right. Draw the bow, the left hand holds the arrow and extends to the left. Turn the head left. Return to the start position and then repeat on the right side. Repeat the whole procedure six times.
Action 3	*Raise one arm to regulate the spleen and stomach* Stand naturally, lift the left hand up and turn the palm upward when it is above the head. Lift hard. Move the right hand downward with the palm facing down. Lift and press for several times and put the hands back. Breathe deeply and force steadily, and repeat on the right side.
Action 4	*Turn the head back to treat consumptive diseases and injuries* Separate the feet to the shoulder width. Hang down the two hands, Calm down and regulate the breath. Sink your breath to the lower abdomen, hang down the two hands, put the palms forward and open downward. Turn the head left slightly and look backward. Repeat on the right side. Repeat six times.
Action 5	*Sway the head and swing the buttocks* Separate the feet, squat the knees, lift when raising the hands and press when laying down, lean right and turn left. Sway the head and swing the buttocks, lean left and turn right. Sway the head and swing the buttocks and repeat six times.
Action 6	*Hold the feet by the hands to reinforce the loins and kidneys* Separate the feet to the shoulder width, raise the arms horizontally over the head. Face the palms forward, lift when raising the hands and press when lay down and bend backward. Being the waist as the axis, bend over slowly and touch your toes.
Action 7	*Punch with anger* Separate the feet, squat the knees, make the hands a fist and put the fist eyes downward. Punch the left fist with anger and return with the hand griped. Repeat with the right hand. Repeat six times.
Action 8	*Shake the spine to help prevent disease* Put the feet together and straight the legs. Relax the body and hang down the arms. Put the fingers together, put the palms forward and lift the toes. Repeat seven times. Relax the whole body.
Closing posture	Fold the two palms at the close of the abdomen, calm down and relax the body, breath evenly and sink your breath to the lower belly.

will not be involved in the subject recruitment process to avoid possible detection bias during data collection and analysis.

Intervention

Baduanjin exercise group

Participants in the Baduanjin exercise group will receive Baduanjin exercise. Participants will be taught and supervised by a professional Baduanjin teacher, including standard requirements for each movement of Baduanjin exercise, breathing techniques, meditation methods and other relevant techniques. The specific exercise programme is in accordance with the Health Qigong Baduanjin Standard published by the General Administration of Sports,[31] and specific action requirements of Baduanjin exercise are shown in table 2. The duration of each exercise will be at least 30 min, roughly between 9 and 10 a.m., and the intervention frequency will be five times a week. About 5 min of warming up will be done before Baduanjin exercise and 5 min of relaxation will be done after Baduanjin exercise. During the exercise, theBaduanjin teacher will supervise, guide and correct the movements of the participants and guide the participants to breathe and meditate to ensure the accuracy of the effect and results of exercise therapy.[32]

The Borg's rating of perceived exertion scale will be used to monitor exercise intensity during the intervention (12–13 scale or 55–75% of the peak HR).[30] Exercise logs for all patients during the intervention will be recorded and kept.

The control group

The control group will continue to receive the standard care,[33] as well as health education on CRF in patients with PCa, including basic knowledge and strategies for self-management of CRF. The instructors providing the

health education are professors from Shanghai University of Traditional Chinese Medicine with over 10 years of experience in diagnosing and treating CRF. The standard care for participants in the control group will comprise antagonistic treatment recommended by the National Comprehensive Cancer Network (NCCN),[34] follow-ups and health education. Participants will be reminded to have regular follow-up, including monitoring of serum testosterone, creatinine, liver function and metabolic parameters at 3 to 6 month intervals. If participants are on long-term ADT, their initial bone mineral density will be measured to assess the fracture risk. Participants will be counselled on the beneficial effects of exercise to avoid ADT-related toxicity. The content of health education is based on CRF: ESMO Clinical Practice Guidelines for diagnosis and treatment.[17] In addition, participants will be encouraged to keep their regular routine. Any special exercise will be recorded. In the future, related data will be used for subgroup analysis of the effects of these exercise on the outcomes. After completing RCT, participants in the control group will have the opportunity to receive Baduanjin exercise teaching.

Outcome measurements and follow-up

This study will evaluate the outcome indicators at three time points: baseline, the end of 12 week intervention and the end of the 12 week follow-up. All outcome measures and follow-ups will be conducted by members of this study group. The study is proposed to be a single-centre, assessor and statistician blinded, RCT. The outcome measurement will include basic information and primary and secondary indicators. The primary time point will be 12 weeks post-intervention. The secondary time point will be 12 weeks after follow-up. The primary indicator will be the PFS, while the secondary indicators include the BFI, PSQI, BDI and FACT-P.[35] After the end of the trial, the mean and SD of the two sets of each outcome indicator will be combined with the corresponding formula to test the power of the trial and judge the reliability of the trial results.

Basic information

Participants' sociolect-demographic data (eg, age, educational background, employment status, marital status and household income) and medical history (eg, date of diagnosis, current stage of PCa and date and type of treatment) will be assessed at baseline (T1). With the patient's informed consent, baseline measurements will be completed before the randomisation.

Primary outcomes

The PFS is a multidimensional fatigue scale that reflects patients' subjective feelings on three different dimensions: somatic, emotional and cognitive. It will be used to assess the severity of fatigue pre- and post-treatment. There are 22 items covering four sub-scales: how much fatigue affects daily activities, how fatigue is affected in terms of emotions and how fatigue is affected in terms of somatic symptoms and cognitive and emotional states, all expressed digitally on a scale of 0–10, where 0 indicates no fatigue and 10 indicates the most severe fatigue. In addition, 1–3 indicates mild fatigue, 4–6 indicates moderate fatigue and 7–10 indicates severe fatigue. The scores for both the total scale and each sub-scale are the sum of the scores of the included items divided by the total number of entries. Overall, higher scores indicate more severe fatigue in patients.[36 37]

Secondary outcomes

Secondary indicators will assess the clinical impact of Baduanjin exercise on the patient's fatigue, sleep, depression and life quality at the time. Measurements will include BFI, PSQI, BDI and FACT-P.

Fatigue: BFI

The BFI is a uni-dimensional assessment scale designed to assess the degree and impact of fatigue in the past 24 hours, covering the impact of fatigue in the past 24 hours on daily behaviour, mood, mobility, daily tasks, relationships with others and enjoyment of life.[38 39] A 10-point numerical scale is used to describe the degree of fatigue, with higher scores associated with higher levels of fatigue. Clinically, 1–3 is considered mild, 4–6 moderate and 7–10 severe. Overall, higher scores can be associated with higher levels of fatigue.

Sleep: PSQI

The PSQI is used to assess the subjective sleep quality of patients. It can be divided into seven factors (sleep quality, time to fall asleep, sleep duration, sleep efficiency, sleep disturbance, sleeping medication and daytime function) and a total score for comparison. Each factor is scored as 0, 1, 2 or 3, and the cumulative score for each factor is the total score. The total PSQI score ranges from 0 to 21, the higher the score, the worse the sleep quality. A score of ≤7 is considered clinically normal and ≥8 is diagnosed as a sleep disorder.[40] The Chinese version of the PSQI is a reliable and valid scale and has been widely used in cancer patients.[41]

Depression: BDI

The BDI is a self-rating scale for determining the degree of depression or depression. The scale is divided into a total of 21 items, including depression, pessimism, feelings of failure, dissatisfaction, feelings of self-guilt, feelings of self-disappointment, negative tendencies, social withdrawal, indecision and distorted body image, difficulties at work, fatigue and decreased appetite. Each entry is scored on a 0, 1, 2 or 3 scale, and the cumulative score for each factor is the total score. The total score range for the BDI-13 is 0–39, the higher the score, the more severe the depression.[42] The BDI, one of the most commonly used self-rating scales for depression, is a classic tool used to screen for depressive states and has shown internal consistency and validity in non-psychiatric patient populations.[43]

Life Quality: FACT-P

The FACT-P is a multidimensional, patient self-administered questionnaire about life quality. The

FACT-P consists of 27 cancer-specific items (FACT-G) plus 12 PC-specific items. The 27 cancer-specific items include four domains: physical health (seven items), social and family well-being (seven items) and emotional well-being (six of the 12-item PC-specific section explored general symptoms, genitourinary and rectal functional status. Each question has a five-item Likert-type scale ranging from 0 = 'not at all' to 4 = 'very much', the higher the score, the higher the patient's approval. According to a recent systematic evaluation, the FACT-P has been included in the assessment of the quality of life in PCa patients in approximately 80% (11/14)[18 44] of cases over the last 10 years and has been proven useful in determining functional status and may also reflect the response of PCa patients to treatment.[45 46]

Data management

Each researcher will receive unified training to ensure that they can collect indicators in a uniform, standard manner. All indicators will be collected peer-to-peer by members in charge of recruitment, and the members will inform all participants and obtain their support during recruitment. Participants will be followed throughout the intervention to prevent shedding.

All researchers will maintain the confidentiality of all participants' personal information and indicator results. Two members accountable for data will enter all collected data into the computer using a double data entry method. Data cleaning will be performed before data analysis to prevent discrepancies or coding errors. The electronic data will be checked against the paper records of the original data. Furthermore, the data entered will then be double-checked for accuracy by two other members. Finally, all electronic data will be stored in a zip folder using a password-protected access system, and hard copies of all material will be stored in a specific cabinet.[47] Data will be periodically audited and verified internally. Data monitoring will be conducted by the Science and Technology Department of Acupuncture and Tuina Institute, Shanghai University of Traditional Chinese Medicine. The monitors will verify that all AEs are recorded in the correct format and as defined by the protocol. The monitors will verify the following variables for all participants: initials, date of birth, gender, signed informed consent, eligibility criteria, date of randomisation, treatment assignment, AEs and endpoints. The administrators will be required to report weekly data monitoring progress to the Steering Committee regularly, including the accuracy and reliability of data.

Data analysis

For this study, all data processing and statistical analysis will be performed by SPSS statistical software version 21.0 (IBM). An intention-to-treat analysis will be used for the management of missing data. Effect sizes for between-group comparisons will be estimated using Cohen's d.[48] Two independent sample t-tests and X^2 tests will be used to assess baseline data such as demographics, disease and treatment as well as differences between the experimental and control groups.[49] We will use repeated-measure analysis of variance to explore how the primary outcome (PFS) and all the other secondary outcomes (BFI, PSQI, BDI and FACT-P) varied over time, between groups, and between time and group interactions. A repeated measures model with two levels of treatment variables (Baduanjin exercise group and the control group) and three times (on enrolment, end of 12 week intervention and end of 12 week follow-up) will be used to test the effectiveness of Baduanjin exercise in reducing fatigue, depression, and improving sleep quality and quality of life. $p<0.05$ (bilateral) was considered statistically significant.[50] Heterogeneity tests will be conducted, and subgroup analysis will be performed for those with high heterogeneity.

DISCUSSION

PCa is considered to be one of the most frequently diagnosed cancers in men on a global scale.[2] In the USA, PCa is the most prevalent male-related cancer, accounting for 29% of new cancer diagnoses in 2022.[1] In the past decade, the incidence and mortality of PCa also has shown a rapid upward trend in China. According to the 2015 National Cancer Registry, PCa ranks as the sixth most common malignancy in men, accounting for 3.35% of malignancy deaths in men.[51] The 5 year survival rate for PCa is 98% at all stages combined,[1 52] highlighting the potential need to deliver survivorship care and understand survivors' health concerns.

PCa is hormone dependent, with the androgen testosterone and dihydrotestosterone responsible for driving disease progression.[53] Hormone treatment may control advanced PCa for long periods by shrinking the size or limiting the growth of the cancer, thus helping to relieve pain and other symptoms.[54] One of the most common treatment modalities for PCa is ADT,[55] with approximately half of patients receiving this therapy over their courses of treatment, either as a primary or adjuvant therapy.[55] The purpose of ADT is to reduce the blood levels of androgens through either surgical or medical interventions, including androgen-targeted therapy.[55]

However, ADT for PCa treatment is associated with adverse physiological changes[5] and various side effects[56] including CRF. CRF remains one of the most common cancer-related symptoms reported by cancer survivors, particularly for PCa survivors undergoing ADT.[8 57–59] Although other symptoms may improve after the end of ADT, CRF persists in a large number of cancer patients. CRF affects the patient's daily life both physiologically and psychologically.[60–62] The main manifestations of CRF are fatigue, inexplicable depression, lethargy and lack of interest in life.[63] At present, there are many studies on the pathogenesis of CRF, among which, the explanation of CRF by 5-hydroxytryptamine (5-HT) disorder has been accepted by most researchers.[64–66] This mechanism can be divided into two types of CRF: peripheral and central

fatigue.[67] Peripheral fatigue mainly refers to physical fatigue. After cancer patients receive chemotherapy, peripheral nerves can be stimulated to release neuroactive substances, and the vagus afferent nerve can be activated, thus inhibiting skeletal muscle activity. A decrease in skeletal muscle activity leads to physical fatigue. Central fatigue, including emotional and cognitive fatigue, is mainly related to neural bundles and disorders in the brain, especially the increased concentration of 5-HT in the brain.[68] Despite ongoing efforts and research in this area, the mechanism of CRF remains unclear. There is no evidence to support the existence of specific intervention drugs. Nonpharmacological methods currently used for CRF alleviation remain challenging.[69 70]

There is an increasing number of patients and physicians preferring complementary and complementary medicine like TMC aerobic exercise to control the symptoms of chemotherapy patients and improve their quality of life.[71] Baduanjin is one of the TMC aerobic exercise therapies of mild to moderate intensity, which is considered as an effective approach to improving health conditions.[72] Baduanjin training intensity fulfilled the American College of Sports Medicine's recommendations to stimulate the body, causing physiologically oriented outcomes.[73] Baduanjin emphasises the connection between the body and mind. It involves relaxing movements while breathing deeply and stretching muscles with mental concentration. It has been shown to have profound therapeutic effects on patients suffering from a variety of ailments.[25]

Baduanjin has the potential to improve CRF and other side effects associated with PCa and ADT treatment.[24] As a non-drug intervention for CRF, Baduanjin exercise can alleviate the degree of CRF in patients and also improves their quality of life.[74] Numerous studies[25 75 76] have shown that Baduanjin exercise is effective in relieving physical pain, improving physical function and relieving negative emotions such as anxiety and depression. Baduanjin exercise training can improve limb motor function, balance, muscle strength and gait function, thus relieving peripheral fatigue.[73] Hypothalamic-pituitary-adrenal, which is involved in activating the immune system and inflammatory response, plays a role in the dysregulation processes that directly or indirectly contribute to the development of CRF.[77] As a kind of low to medium-intensity aerobic exercise,[78] Baduanjin exercise can stimulate the pituitary gland to secrete beta-endocannabinoids, improve central nervous system responsiveness and body tolerance, relieve muscle tension and mental depression through micro-electrical stimulation of the nervous system and ease the cerebral cortex, thus delaying the production of fatigue.[79] Baduanjin exercise improves anxiety in patients with generalised anxiety disorder[80] and type 2 diabetics.[81] Studies have reported specific changes in protein expression in patients with mental mood changes. Wang Qingxin *et al* used SELDI[82] to identify serum-associated anxiety proteins in the serum of tumour patients. Han Yan *et al* found that the positive detection rate of serum-associated anxiety proteins in the intervention group was significantly lower than that in the control group at 3 months of Baduanjin training, suggesting that Baduanjin training is beneficial in relieving anxiety in postoperative breast cancer patients.[83]

As with all related studies, this trial has a few potential limitations. Like any other behavioural intervention trial, this trial has a few potential limitations on blinding. However, the outcome assessors and statisticians will be blinded to the intervention allocation. Given the limited study sites, the sample size of this trial is small, which will contribute to only a preliminary analysis of the effects of Baduanjin exercise therapy on the CRF of PCA patients under ADT treatment. However, our results can only be considered suggestive; results require replication in larger, more diverse populations, preferably in a randomised trial.

To our knowledge, this is the first study to investigate the efficacy of Baduanjin exercise alone in managing prostate CRF during and after the ADT treatment.[7] Several studies have been completed[84 85] or designed[86] to investigate the positive outcomes of exercise in improving CRF. As a low-to-moderate-intensity exercise, Baduanjin exercise can lead to similar health improvements in high-risk populations with better adherence, lower injury risk and long-term sustainability than conventional exercise.[87–90] The results of this trial will contribute to a future multicentre large-scale main RCT to further conclude the research evidence on the effects and of Baduanjin for CRF management in PCa patients under ADT treatment.

Ethics and dissemination

This study protocol (v1.1, 14/08/2023) has been approved by Shanghai Hudong Hospital Ethics Committee, Shanghai province (2022 SHHDKY08). The study protocol has been registered and can be obtained through the Chinese Registry website (registered in ChiCTR.org with the identifier), and the trial registration number was ChiCTR2300074293. If patients experience adverse reactions, they will receive the corresponding treatment. The results of this study will be presented at local, national or international conferences on CRF. It will be submitted as a manuscript to a peer-reviewed journal. The main results of the study will also be shared with all participants. It will also be disseminated through courses, lectures and the internet to researchers, healthcare providers, healthcare professionals and the general public, regardless of the size or direction of the impact.

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Contributors GG is the guarantor. YW and GG planned the study protocol and drafted the manuscript. YW, XX, XZ, GY, YX, GG, YQ, LL and FY planned the study protocol and revised the manuscript. GG Yijinjing exercise group will carry out the Yijinjing exercise intervention of patients and will be responsible for generating and distributing the random numbers. YQ health education group will carry out the health education of patients, and will recruit and screen eligible participants in the outpatient department and the in-patient department and will participate in designing the outcome measurements and assessing the outcomes. XX was the study coordinator. LL, FY and GG managed the study. All authors read and approved the final manuscript.

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